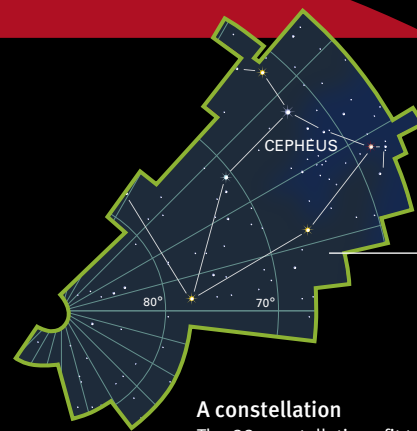
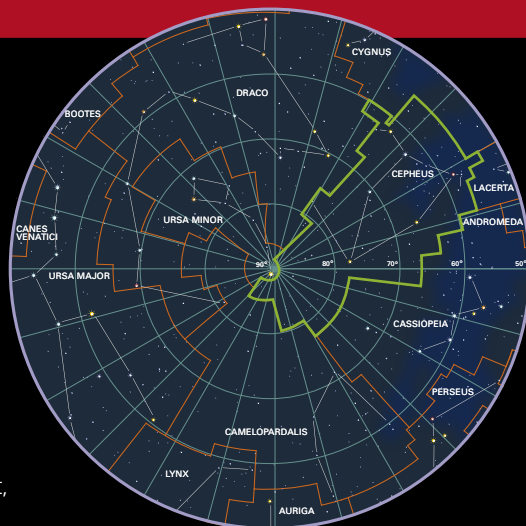


DIVIDING THE SKY

At first glance, stars seem like indistinguishable pinpoints of light randomly scattered across the sky. Quite quickly, however, it is noticeable that some stars are brighter than others, and when these are linked together, recognizable patterns emerge. Astronomers have been using such patterns, the constellations, for about 4,000 years to navigate around the night sky. Today, the sky surrounding Earth is divided into 88 internationally recognized constellations, just over half of which depict mythological people and creatures. The constellation Orion depicts the eponymous Greek hunter; the constellation Scorpius is the scorpion that killed him. They vary in size and complexity. Hydra, the water snake, is the largest, and Crux, the southern cross, the smallest.



constellation Cepheus represents mythical King Cepheus of Ethiopia

A constellation

The 88 constellations fit together to form the celestial sphere that surrounds Earth. Each constellation consists of a star pattern and the sky immediately around it. Cepheus (above) can be seen by all northern-hemisphere observers.

THE CHANGING SKY

STARRY BACKDROP

Some of the first stars to be grouped into recognizable patterns were those that form the background to the Sun as it moves across the sky. The Sun's path against the backdrop of stars is called the ecliptic. And the band of sky centered on the ecliptic is known as the zodiac.

CIRCLE OF ANIMALS

The name zodiac comes from the Greek for animal, and it refers to the patterns made by the stars within this band of sky. With the exception of Libra, which is a set of scales, the traditional zodiac is a group of creatures. The zodiac constellations are, in order around the sky: Aries, Taurus, Gemini, Cancer, Leo, Virgo, Libra, Scorpius, Sagittarius, Capricornus, Aquarius, and Pisces.

The Sun completes one circuit of its ecliptic each year. It takes approximately one month to move through each of the zodiac constellations. However, this is not the full picture. Traditionally, the zodiac is described as consisting of 12 constellations. There is, though, a 13th, Ophiuchus, and the Sun spends more time crossing this constellation than it does traversing its neighbor Scorpius.

The zodiac band of sky also forms the starry backdrop to the planets and the Moon. Their paths take them close to the ecliptic—sometimes to its north, and sometimes to its south.

The zodiac

The band of the celestial sphere that forms the backdrop to the movement of the Sun, Moon, and planets is called the zodiac. It is centered on the ecliptic, the path that the Sun traces out month by month as Earth orbits around it. There are 13 constellations in the zodiac: the 12 traditional zodiacal constellations and Ophiuchus.

